

Self-Check Questions: IEEE-754 Floating Point

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1 IEEE-754 Floating Point

1.1 Basic Structure

1. What are the three components of an IEEE-754 floating point number?
2. What is the purpose of the sign bit?
3. How is the exponent stored (biased or unbiased)?
4. What is the mantissa (significand), and what does it represent?
5. Write the general formula for a normalized floating point number.
6. For single precision (32-bit), how many bits are allocated to:
 - sign
 - exponent
 - fraction
7. For double precision (64-bit), how many bits are allocated to each field?

1.2 Exponent and Bias

1. What is the exponent bias in single precision?
2. What is the exponent bias in double precision?
3. Why is a bias used instead of signed exponent representation?
4. How do you recover the true exponent from the stored exponent?
5. What exponent value corresponds to zero after bias correction?

1.3 Normalization

1. What does it mean for a floating point number to be normalized?
2. Why is there an implicit leading 1 in normalized numbers?
3. Write the formula for a normalized IEEE-754 number including the implicit bit.
4. What happens when the exponent field is all zeros?
5. What happens when the exponent field is all ones?

1.4 Special Values

1. How is zero represented in IEEE-754?
2. Why are there two representations of zero?
3. How is infinity represented?
4. How is NaN (Not a Number) represented?
5. What is the difference between quiet NaN and signaling NaN?

1.5 Subnormal Numbers

1. What are subnormal (denormal) numbers?
2. Why do subnormal numbers exist?
3. How is their representation different from normalized numbers?
4. What is the smallest positive subnormal number in single precision?
5. What is the trade-off when using subnormal numbers?

1.6 Precision and Rounding

1. What is machine epsilon?
2. How many significant decimal digits does single precision provide?
3. How many significant decimal digits does double precision provide?
4. What are the common IEEE-754 rounding modes?
5. What does “round to nearest, ties to even” mean?

1.7 Range and Limits

1. What is the largest representable finite number in single precision?
2. What is the smallest positive normalized number?
3. What is the smallest positive subnormal number?
4. How does exponent size affect range?
5. How does mantissa size affect precision?

1.8 Conversion and Interpretation

1. Convert a decimal number to IEEE-754 format step by step.
2. Convert an IEEE-754 bit pattern to decimal.
3. How do you interpret the exponent and mantissa fields together?
4. What mistakes are common when converting floating point numbers?

1.9 Arithmetic and Errors

1. Why is floating point addition not associative?
2. Give an example where $(a + b) + c \neq a + (b + c)$.
3. What is rounding error?
4. What is catastrophic cancellation?
5. Why do floating point errors accumulate in iterative computations?

1.10 Edge Cases and Pitfalls

1. Why can 0.1 not be represented exactly in binary floating point?
2. What happens when overflow occurs?
3. What happens when underflow occurs?
4. Why should you avoid direct equality comparison of floats?
5. How do you safely compare floating point numbers?

1.11 Implementation Understanding

1. How would you extract the sign, exponent, and mantissa from a 32-bit integer?
2. How can bit manipulation be used to inspect a float in memory?
3. What is the difference between float and double in terms of memory and precision?
4. When should you prefer double over float?
5. When might floating point be inappropriate (e.g., finance)?